MCU

File: mcu.kicad_sch

Power

File: power.kicad_sch

Outputs

File: outputs.kicad_sch

Inputs

File: inputs.kicad_sch

Connectors and CAN

File: connectors.kicad_sch

Wideband controller

File: wideband.kicad_sch

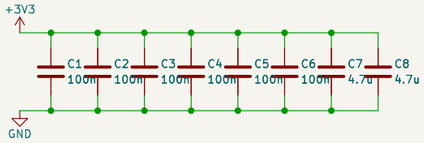
Miata ECU daughterboard rev 0.8 (on-board wideband controller, flashed through the main MCU; 5 V low-side drive, fused logic supply). Block sheets connect through global labels; GND, +3V3, +5V and +12V are power sym

Sheet: /
File: miata_ecu_daughter.kicad_sch

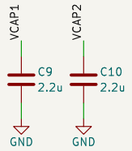
Title: Miata ECU daughterboard

Size: A4	Date: 2026-11-03	**Rev: 0.8**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/7

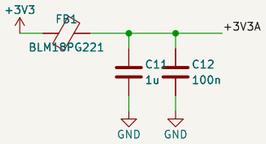
Decoupling: one 100 nF per VDD pin, 2 x 4.7 uF bulk



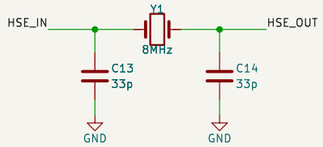
Core regulator caps



Analog supply filter

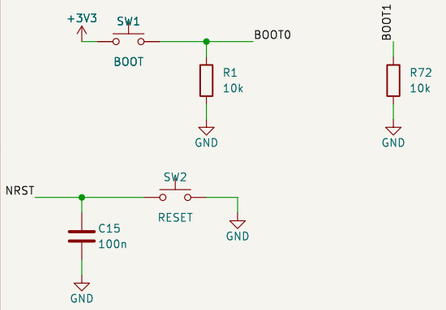


8 MHz HSE crystal

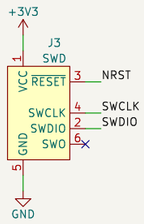


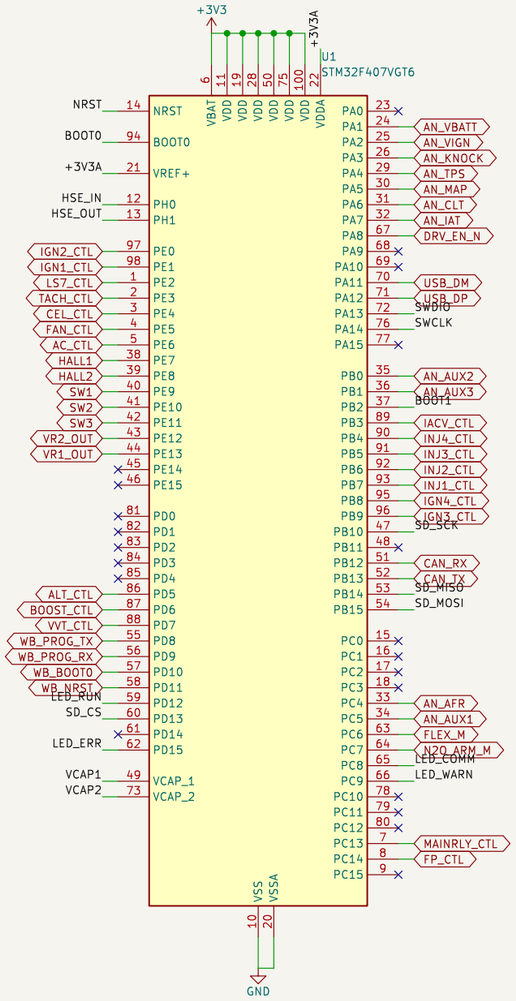
BOOT0 (hold for DFU) and reset

BOOT1 (PB2) low: DFU needs BOOT0 = 1, BOOT1 = 0

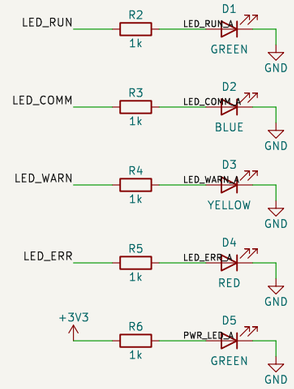


SWD (Tag-Connect TC2030-NL pads)

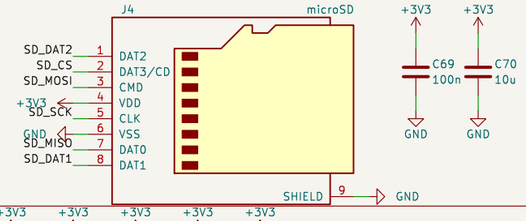




Status LEDs



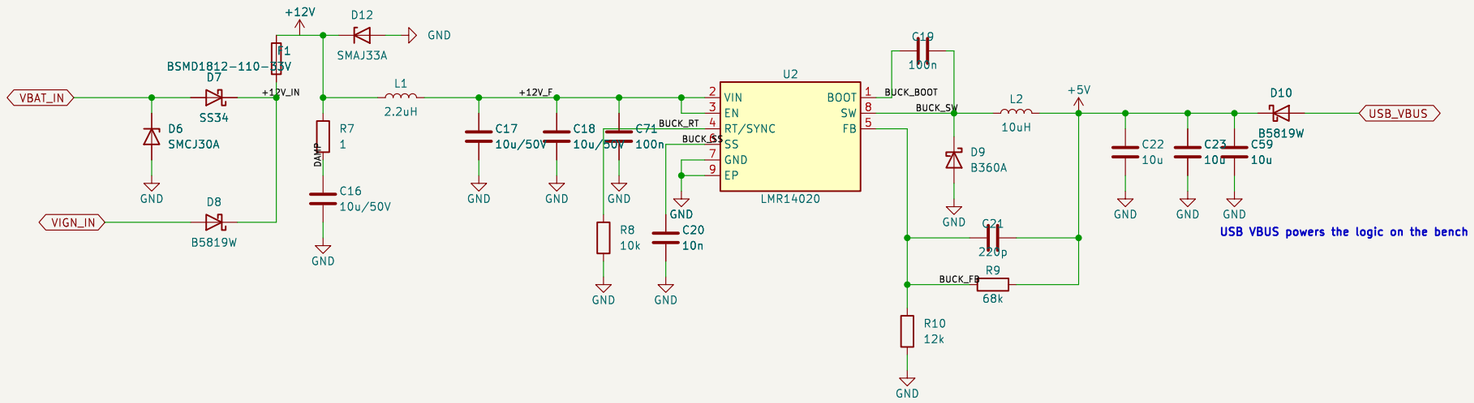
rev 0.10: microSD socket on SPI2 (hinged Molex 47219-2001); pull-ups on every card data line



Generated from scripts/design.py; netlist identical to the routed board

Sheet: /MCU/	
File: mcu.kicad_sch	
Title: Miata ECU daughterboard — MCU	
Size: A3	Date: 2026-11-01
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	**Rev: 0.6**
	Id: 2/7

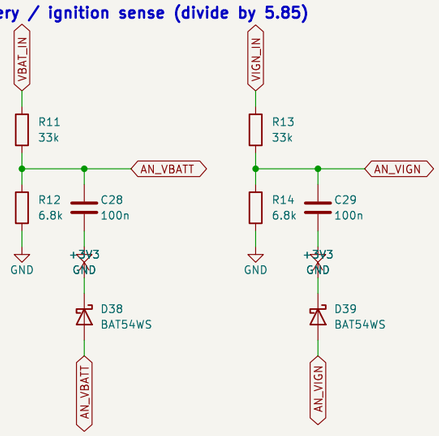
Input protection (fused, TVS-clamped logic supply) and 5 V buck (LMR14020, 5 V / 2 A)



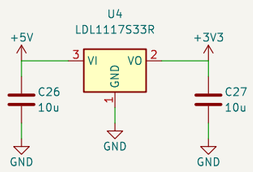
Fsw ~ 1 MHz (RT 10k); Vout = 0.75 V x (1 + 68k/12k) = 5.0 V

USB VBUS powers the logic on the bench

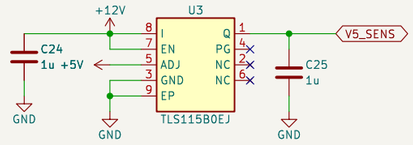
Battery / ignition sense (divide by 5.85)



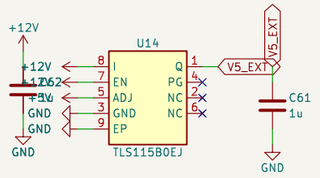
3.3 V LDO



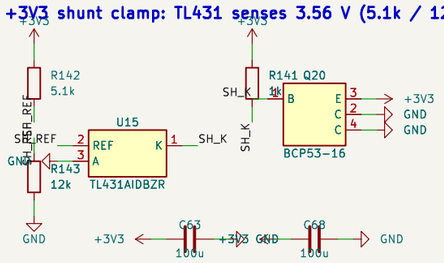
Protected 5 V sensor supply (TLS115 tracks +5V)



Harness sensor 5 V (V5_EXT, second TLS115; J1-13 to the harness and AUX header)



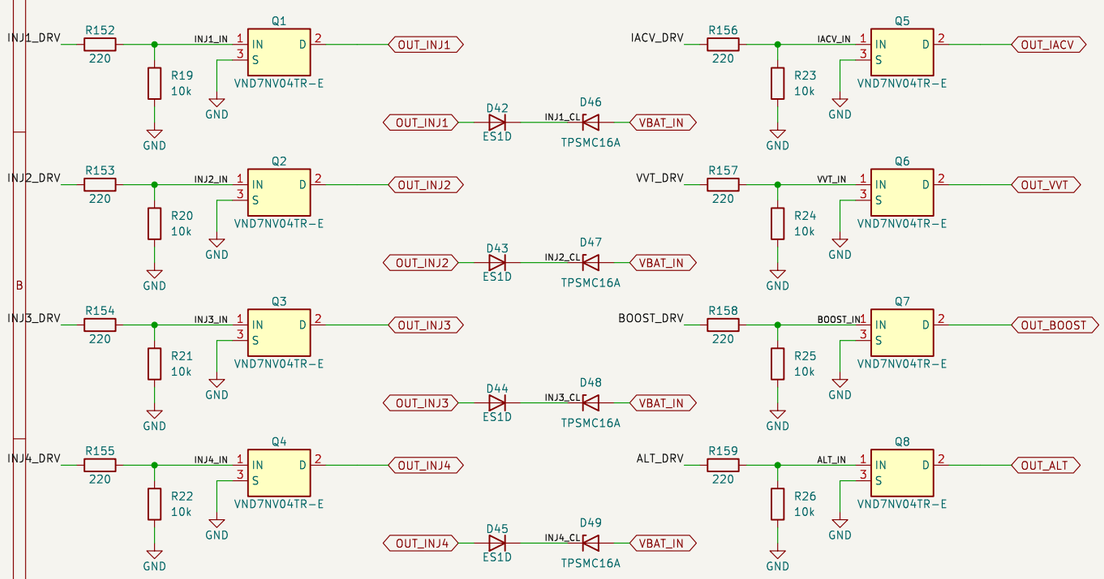
+3V3 shunt clamp: TL431 senses 3.56 V (5.1k / 12k) and turns on a PNP to ground



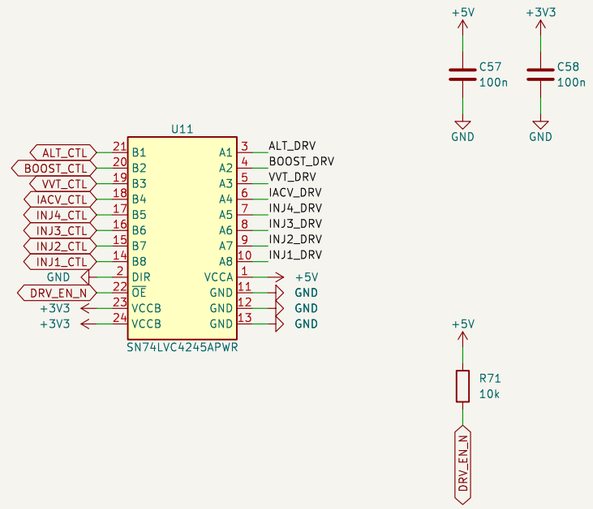
Generated from scripts/design.py; netlist identical to the routed board

Sheet: /Power/ File: power.kicad_sch		
Title: Miata ECU daughterboard — Power		
Size: A3	Date: 2026-11-01	**Rev: 0.6**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/7

Protected low-side drivers (VND7NV04, DPAK): injectors, IACV, VVT, boost, alternator

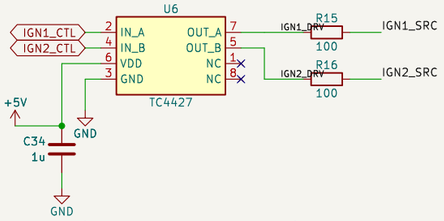


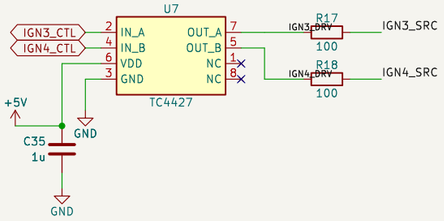
3.3 V -> 5 V level shifter for the low-side inputs (DIR low: B -> A)



-OE pulled up: all eight drivers off until the firmware drives DRV_EN_N low

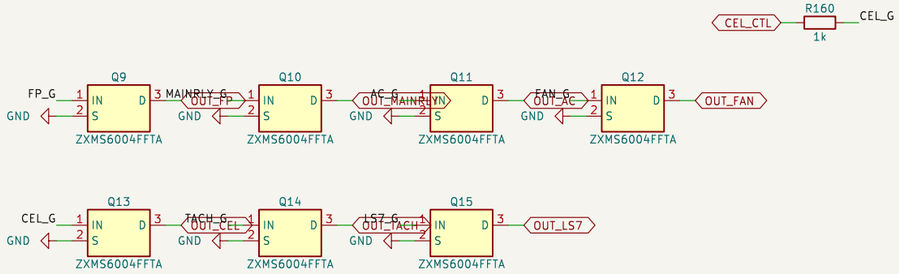
Ignition logic outputs: TC4427 sources through 100 R + Schottky; ZXMS6004 pulls low (gate = inverted control)

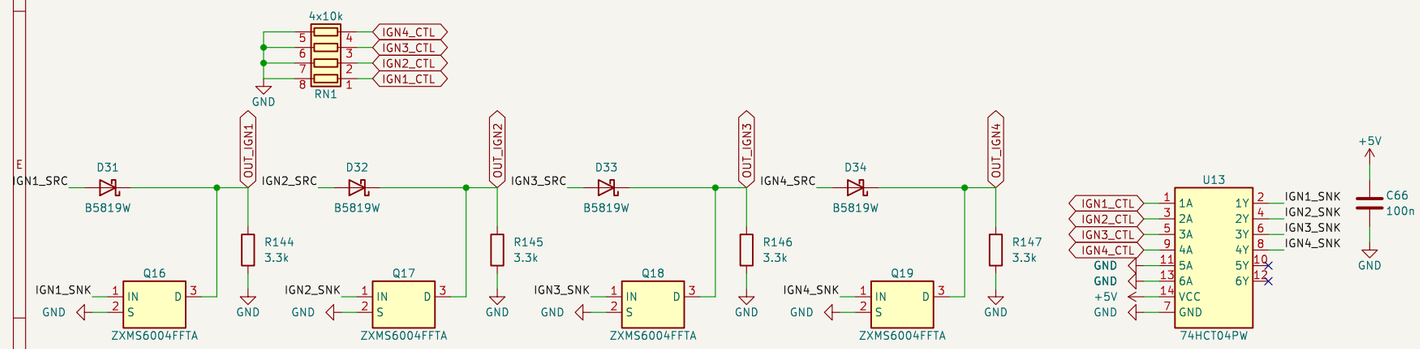


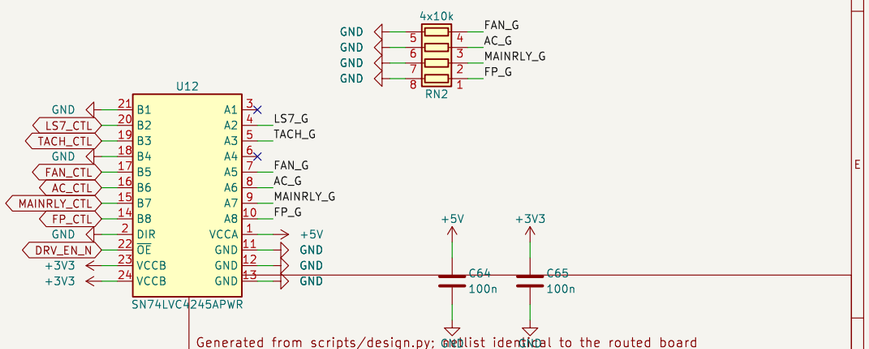


Ignition inputs pulled low (coils stay off while the MCU boots)

Relay, lamp and tach outputs: ZXMS6004 protected low sides, 5 V gates from U12 (enabled with the drivers)



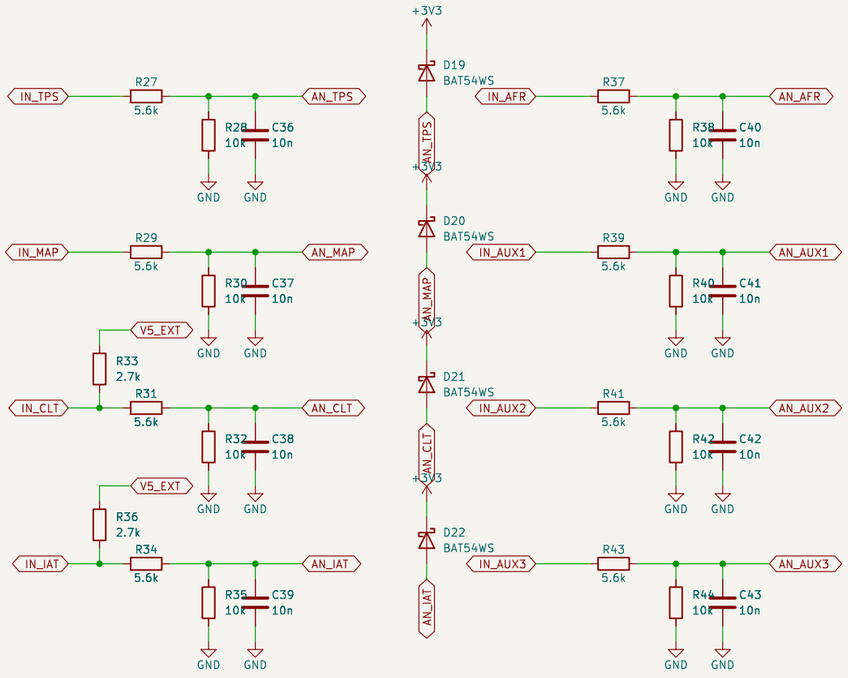




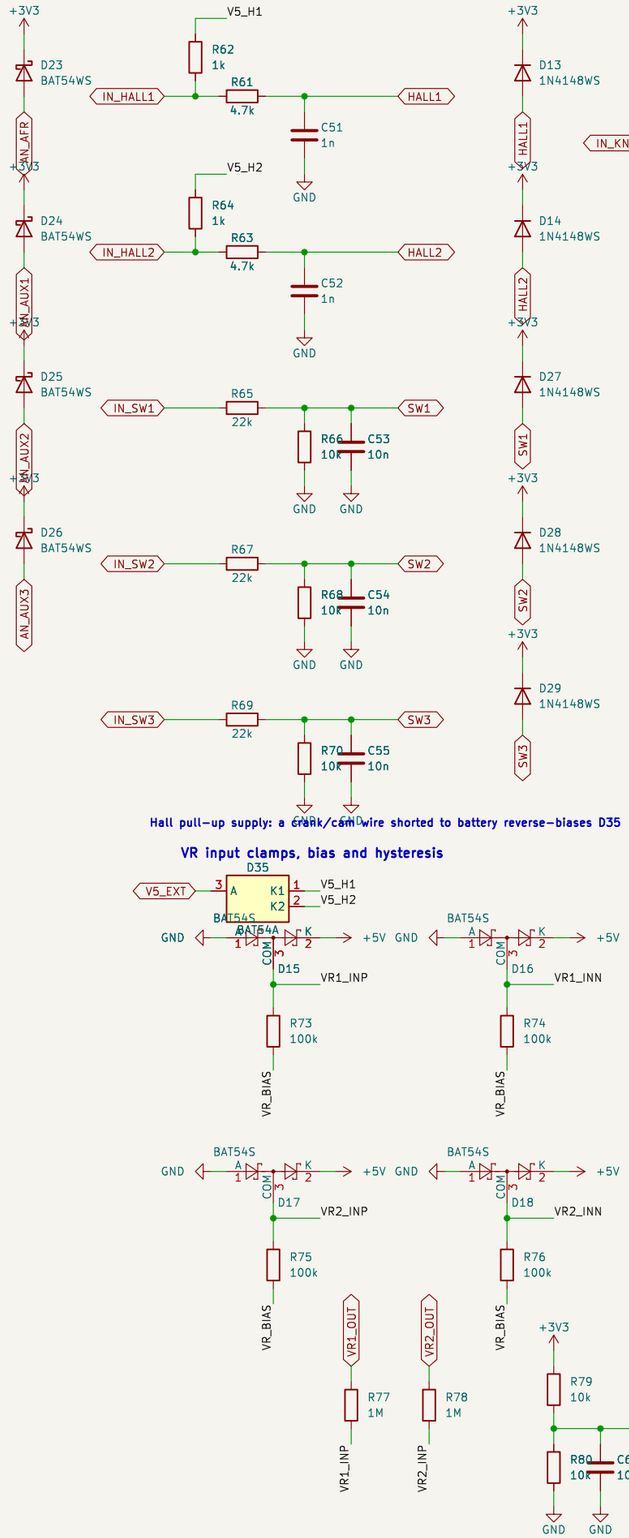
Generated from scripts/design.py; netlist identical to the routed board

Sheet: /Outputs/		
File: outputs.kicad_sch		
Title: Miata ECU daughterboard — Outputs		
Size: A3	Date: 2026-11-01	**Rev: 0.6**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/7

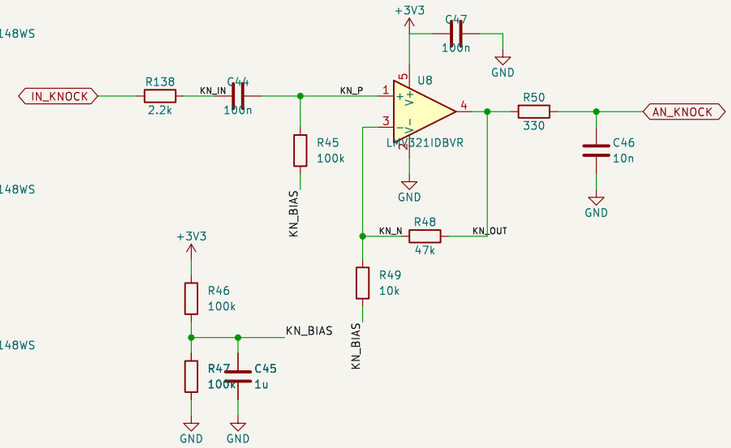
Analog sensor inputs: 0–5 V -> 0–3.2 V for the ADC (5.6k / 10k, 10 nF)



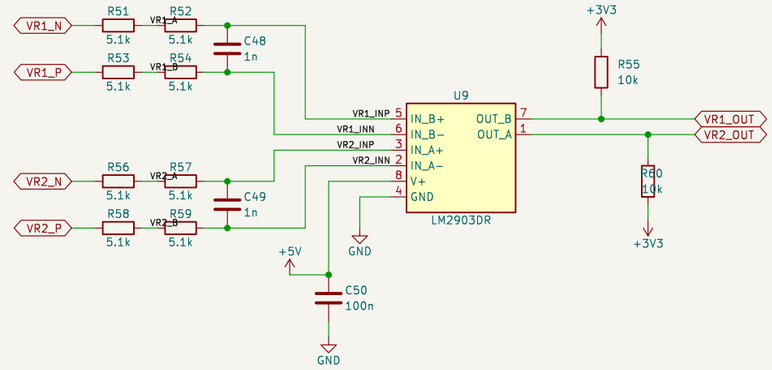
Hall (open collector, 1k pull-up) and 12 V switch inputs



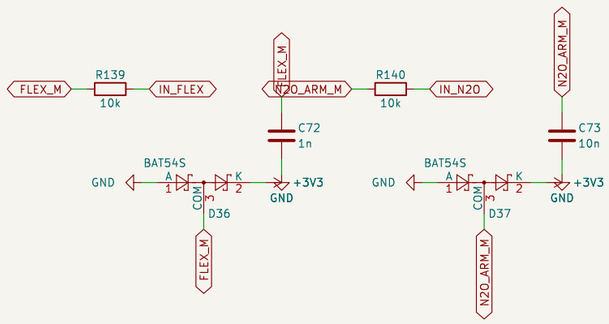
Knock: AC-coupled, biased to 1.65 V, gain 5.7



Crank / cam VR inputs (spare): LM2903 comparator, 1M hysteresis, BAT54S clamps, biased to 1.65 V

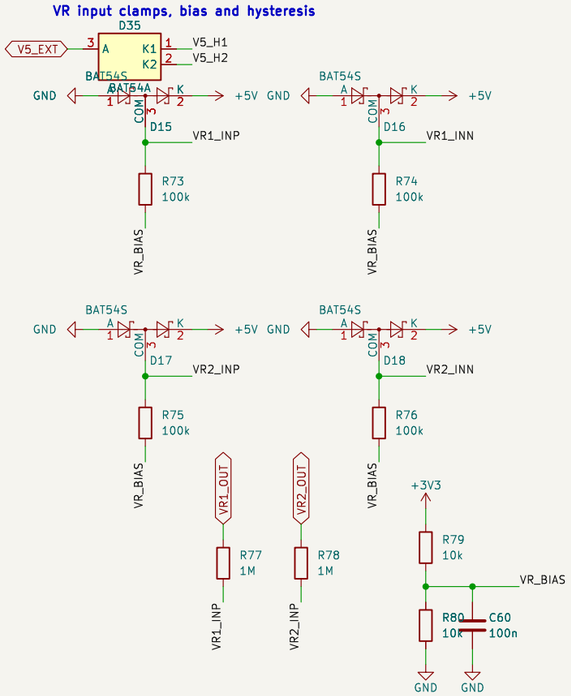


AUX header UART (J2 24/25): 10k series, BAT54S clamps to GND / +3V3



Hall pull-up supply: a crank/cam wire shorted to battery reverse-biases D35

VR input clamps, bias and hysteresis



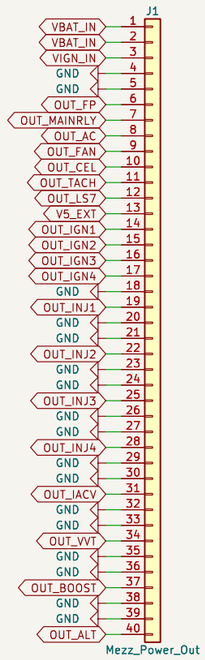
Generated from scripts/design.py; netlist identical to the routed board

Sheet: /Inputs/
File: inputs.kicad_sch

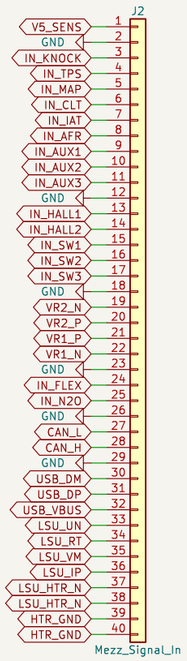
Title: Miata ECU daughterboard — Inputs

Size: A3	Date: 2026-11-01	**Rev: 0.6**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 5/7

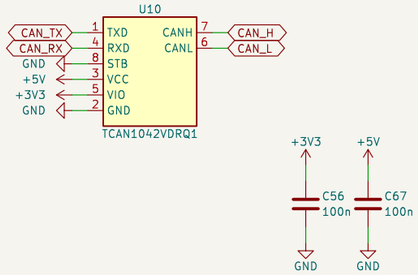
J1: power in and outputs (top edge)



J2: sensor inputs and comms (bottom edge)

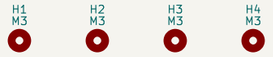


CAN transceiver (termination on the motherboard)



rev 0.8: TCAN1042V, 5 V VCC (C67), 3.3 V VIO (C56), STB low

Mounting holes



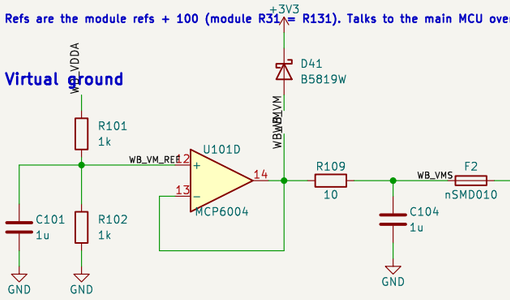
Generated from scripts/design.py; netlist identical to the routed board

Sheet: /Connectors and CAN/ File: connectors.kicad_sch		
Title: Miata ECU daughterboard — Connectors and CAN		
Size: A3	Date: 2026-11-01	**Rev: 0.6**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/7

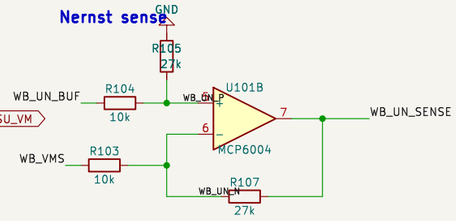
Wideband O2 controller: FOME / rusEFI open-source wideband module (MIT), rev C circuit, laid out on this board

Refs are the module refs + 100 (module R31 = R131). Talks to the main MCU over the shared CAN bus.

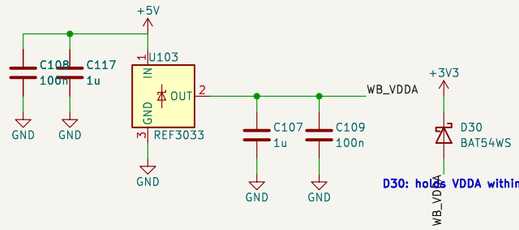
Virtual ground



Nernst sense

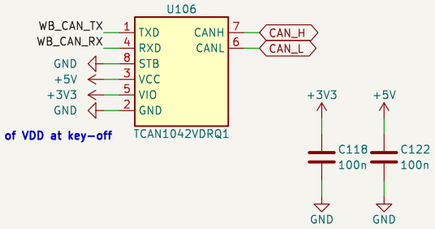


ADC / analog reference

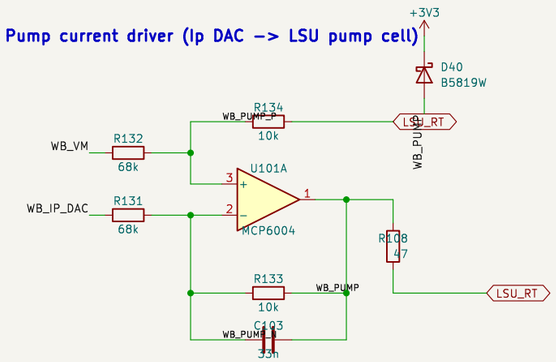


D30: holds VDDA within a diode drop of VDD at key-off

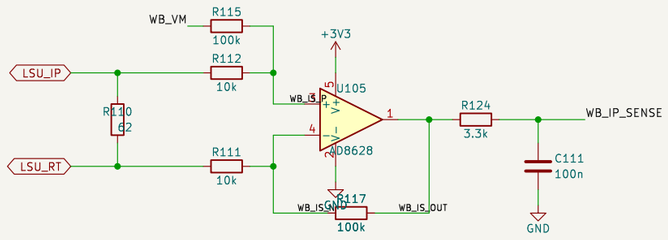
CAN (same bus as the main MCU; rusEFI reads lambda over CAN)



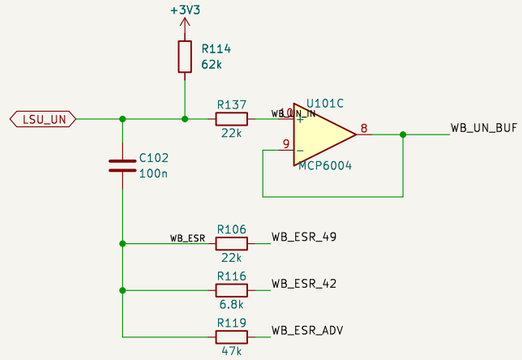
Pump current driver (Ip DAC -> LSU pump cell)



Pump current sense

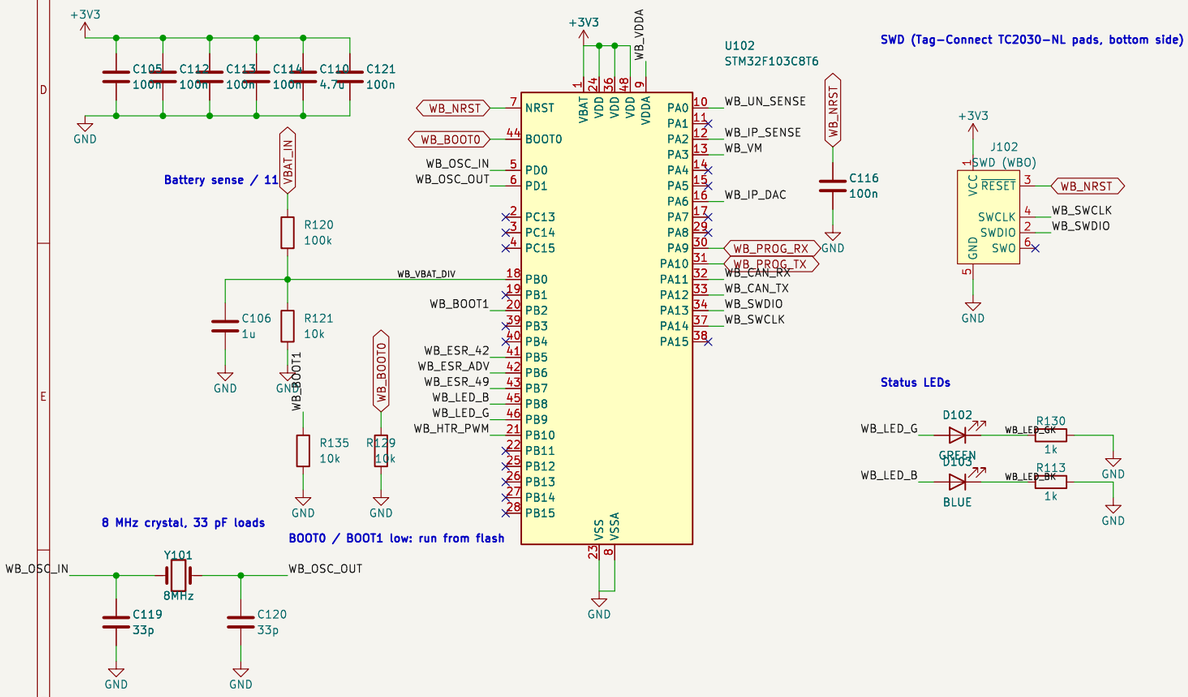


Nernst buffer, bias current and cell resistance (ESR) drive

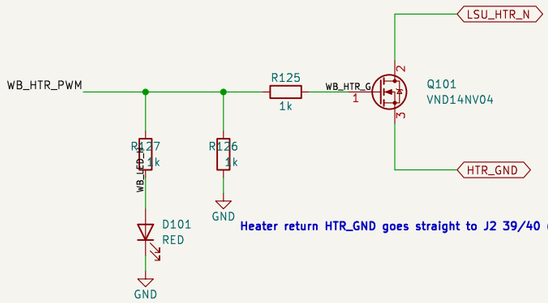


ESR drive: R106 22k / R116 6.8k / R119 47k, through C102 into the Nernst cell

Controller MCU (STM32F103C8: rusEFI wideband firmware, board port in firmware/wideband)



LSU heater low side (protected OMNIFET)



Heater return HTR_GND goes straight to J2 39/40 (motherboard GND), not through this board's plane

Op-amp supply



Generated from scripts/design.py; netlist identical to the routed board

Sheet: /Wideband controller/		
File: wideband.kicad_sch		
Title: Miata ECU daughterboard — Wideband controller		
Size: A3	Date: 2026-11-01	**Rev: 0.6**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/7